QP CODE: 25901795

Scheme of Valuation

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

SECOND SEMESTER MGU-UGP (HONOURS) REGULAR / IMPROVEMENT / REAPPEARANCE EXAMINATION , APRIL 2026

(2024 Admission onwards)

Multi-Disciplinary Course (MDC)

MG2MDCSTA100 – TIME SERIES METHODS AND THEIR APPLICATIONS

Part A -Multiple Choice Questions [Answer **all** Questions. Each Question carries **2 marks]**

1. a) A random signal with no predictable pattern.
2. a) To separate trend, seasonality, and randomness.
3. a) A random walk depends on past values, while white noise does not
4. a) A graphical representation of data points over time.
5. a) No autocorrelation
6. b) Makes the series stationary by removing trends
7. c) -1 and +1
8. d) Weak correlation
9. a) The number of lagged values used
10. a) Differencing the series

[10x2 = 20 marks]

Part B [Answer **any 3** Questions. Each Question carries **5 marks]**

11. Definition/ Explanation of cyclical and seasonal components –2 marks each
    example of cyclical and seasonal component - 1 mark (0.5marks each)
12. Definition of deterministic time series and non-deterministic time series --2.5 marks each

    Or

    giving examples of deterministic time series and non-deterministic time series ---1.5 marks each (maximum of 3 marks)
13. Explaining the idea of semi average method either by words or by example-5 marks.
14. Definition of correlation ----2.5 marks,
    Definition of covariance ---2.5 marks
15. Definition or correct example of Strict sense and weak stationarity process ---2.5 marks each

[5x3 = 15 marks]